A Comparative Analysis of Deep Learning Pipelines for Real-Time Indonesian ALPR and Vehicle Tax Inquiry

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Abstract

The manual verification of vehicle tax compliance (*Pajak Kendaraan Bermotor*) in Indonesia is a labor-intensive process, currently limited to sporadic police checkpoints that disrupt traffic flow. This project proposes a scalable, automated solution: an Automatic License Plate Recognition (ALPR) system integrated with the Banten Provincial Government's live tax database. We engineered and compared two deep learning pipelines to identify the optimal architecture for detecting Indonesian license plates in unconstrained environments: Pipeline A, utilizing the CNN-based YOLOv8, and Pipeline B, utilizing the Transformer-based RT-DETRv2. For the subsequent recognition task, both pipelines employ a novel approach using a fine-tuned Vision-Language Model (VLM), DeepSeek-OCR, to transcribe plate numbers. The models were fine-tuned on a comprehensive dataset of 3,246 Indonesian license plate images. Our results analyze the trade-off between the inference speed of the detectors and the high semantic accuracy of the VLM-based OCR. The final system demonstrates end-to-end capability, successfully detecting plates, extracting alphanumeric strings, and retrieving live tax status, offering a viable proof-of-concept for smart city infrastructure.

Index Terms—*ALPR, YOLO, RT-DETR, DeepSeek-OCR, Indonesia Vehicle Tax, Computer Vision*

1 Introduction

The rapid urbanization of Indonesia has led to an explosive growth in vehicle ownership, particularly in densely populated provinces like Banten. As of 2025, the number of registered vehicles continues to rise, placing a significant strain on administrative infrastructure and compliance monitoring mechanisms. Vehicle tax (*Pajak Kendaraan Bermotor* or PKB) constitutes a critical revenue stream for regional governments, funding essential infrastructure and public services. However, tax compliance remains a persistent challenge. Data from the Ministry of Finance indicates that the national tax ratio is projected to fluctuate around 10.03% in 2025 [1], highlighting a significant gap between potential and realized revenue.

Currently, the primary method for enforcing vehicle tax compliance relies on manual police checkpoints, locally known as *razia*. These operations involve law enforcement officers physically flagging down vehicles to inspect their registration documents (STNK). This manual approach suffers from several critical systemic flaws:

1. **Inefficiency and Scalability:** Officers can only inspect a tiny fraction of vehicles on the road at any given time, leaving the vast majority of non-compliant vehicles undetected.
2. **Traffic Disruption:** Checkpoints often cause significant congestion, creating bottlenecks on major thoroughfares.
3. **Human Error:** Manual verification is prone to fatigue and human error. Furthermore, the arbitrary nature of spot checks can lead to allegations of selective targeting [2].
4. **Lack of Real-Time Integration:** Officers often lack immediate access to a centralized database, requiring time-consuming manual lookups.

To address these challenges, there is an urgent need for an automated, non-intrusive solution capable of monitoring compliance at scale. This project proposes the development of a high-performance Automatic License Plate Recognition (ALPR) system specifically engineered for the Indonesian context.

A core technical challenge in realizing this system is the accurate recognition of Indonesian license plates. Unlike standardized European or American plates, Indonesian plates exhibit significant

variability. They feature diverse background colors (black, yellow, red, white), varying fonts, and often suffer from physical degradation or non-standard modifications [3]. Furthermore, environmental factors such as tropical rain, strong shadows, and the chaotic nature of mixed traffic introduce noise that traditional Optical Character Recognition (OCR) systems struggle to handle.

The contributions of this work are threefold:

- **Comparative Analysis:** A rigorous performance benchmark of CNN vs. Transformer architectures (YOLOv8 vs. RT-DETRv2) for detecting Indonesian license plates.
- **VLM Integration:** The novel application of the DeepSeek-OCR Vision-Language Model to the domain of Indonesian ALPR, identifying its ability to resolve semantic ambiguities.
- **Civic Tech Prototype:** The development of an end-to-end software pipeline connecting computer vision outputs directly to a live government tax API.

2 Related Works

The field of ALPR has evolved significantly, transitioning from traditional image processing techniques to sophisticated deep learning architectures. This section reviews existing literature on object detection, optical character recognition (OCR), and the emerging shift toward multimodal models.

2.1 Object Detection: CNNs vs. Transformers

For the past decade, Convolutional Neural Networks (CNNs) have been the de facto standard for object detection. The YOLO (You Only Look Once) family has dominated the field due to its balance of speed and accuracy. Putra and Yohannes [4] demonstrated the effectiveness of YOLOv8 in detecting Indonesian license plates, achieving high precision even in complex traffic scenes. YOLOv8 introduces an anchor-free detection head and a decoupled architecture, which simplifies the training process and improves generalization across different object sizes [5].

However, CNNs struggle with capturing global context due to the limited receptive field of their convolutional kernels. This limitation can lead to failures in crowded scenes where occlusion is common. To address this, the RT-DETR (Real-Time Detection Transformer) architecture was introduced. Lv et al. [6] presented RT-DETRv2, which utilizes self-attention mechanisms to model long-range dependencies. Benchmarks suggest that RT-DETRv2 can outperform YOLO models in complex scenarios by better handling occlusions, albeit often at a higher computational cost.

2.2 Optical Character Recognition (OCR)

Traditional ALPR systems typically employ a two-step recognition process: character segmentation followed by classification. Methods like LPRNet [7] and CRNN (Convolutional Recurrent Neural Network) [8] have been widely used. LPRNet offers an end-to-end approach that is lightweight and fast, making it popular for real-time applications. However, these models often require perfectly rectified images and struggle with the "in-the-wild" distortions. Research on Indonesian plates [3] noted that standard OCR engines often misinterpret local fonts—specifically confusing 'B' with '8' or 'D' with '0'—due to a lack of contextual awareness.

2.3 The Shift to Vision-Language Models (VLMs)

The integration of linguistic understanding into optical recognition marks a significant paradigm shift. Traditional systems treated ALPR strictly as a pattern matching problem. However, the

release of **DeepSeek-OCR** in October 2025 [9] capitalizes on the advancements of Vision-Language Models to treat license plate recognition as a semantic generation task.

Early Transformer-based OCR: Prior to 2025, the shift began with models like TrOCR [10], which replaced the CNN-RNN-CTC backbone with an end-to-end Transformer architecture. By coupling a Vision Transformer (ViT) encoder with a text-decoder, TrOCR demonstrated that models could learn "language priors." Similarly, PARSeq [11] utilized attention mechanisms to refine character predictions based on bi-directional context. While effective, these models often lacked the spatial awareness required for the rigid three-component structure of Indonesian plates.

The DeepSeek-OCR Advantage: As noted in a survey by Zhang et al. [12], the newest generation of VLMs bridges this gap. DeepSeek-OCR utilizes "Contexts Optical Compression" to retain high semantic understanding while drastically reducing parameter count. By treating the license plate string as a semantic sequence, the model resolves local ambiguities not just by pixel shape, but by understanding the syntax of Indonesian plates (e.g., a plate cannot end with a number if it follows the Regional Code pattern).

3 Methodology

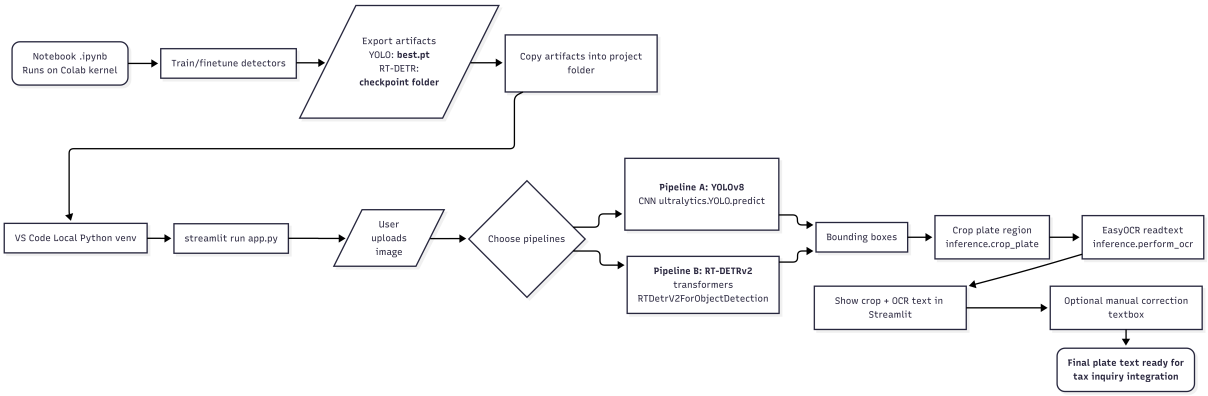

Figure 1: Project Methodology

3.1 Dataset Description

The project utilizes the **Indonesian Plate Number from Multi-Sources** dataset, publicly available on Kaggle. This dataset is structured into two distinct components tailored for our two-stage pipeline:

- **Plate Detection Dataset:** Contains 1,383 images with COCO-format annotations (bounding boxes). This subset is used to train the YOLOv8 and RT-DETRv2 detectors to localize plates within full vehicle images.
- **Plate Text Dataset:** Contains 1,863 cropped license plate images with corresponding text labels. This subset is used to fine-tune the DeepSeek-OCR model to recognize the specific alphanumeric format of Indonesian plates (Regional Code - Number - Sub-region Code).

Data splitting was performed with an 80-10-10 ratio for training, validation, and testing, respectively.

3.2 Pipeline Architectures

We evaluate two distinct pipelines to determine the optimal balance between speed and accuracy.

3.2.1 Pipeline A: CNN Baseline (YOLOv8 + DeepSeek-OCR)

This pipeline utilizes YOLOv8 (You Only Look Once v8) for the detection stage. YOLOv8 employs a CSPDarknet53 backbone and a Path Aggregation Network (PANet) neck to extract features at multiple scales. It is selected for its proven real-time inference capabilities on edge devices.

3.2.2 Pipeline B: Transformer Baseline (RT-DETRv2 + DeepSeek-OCR)

This pipeline utilizes RT-DETRv2 (Real-Time Detection Transformer v2). Unlike YOLO, RT-DETRv2 uses a hybrid encoder and a transformer decoder with self-attention mechanisms. This allows the model to process global image context, theoretically improving detection performance in cases of partial occlusion or complex backgrounds.

3.2.3 Recognition Module: DeepSeek-OCR

Both pipelines feed the cropped license plate image into DeepSeek-OCR. Unlike traditional OCR that segments characters, DeepSeek-OCR utilizes a visual encoder to map the image into token embeddings, which are then processed by a Large Language Model (LLM) decoder to generate the text sequence.

- **Fine-tuning Strategy:** We utilized Low-Rank Adaptation (LoRA) via the `unsloth` library to fine-tune the model efficiently on a T4 GPU. This allows the model to adapt to the specific font styles of Indonesian plates without retraining the entire parameter set.
- **Prompt Engineering:** The model is prompted with the instruction: *"Read the Indonesian license plate number exactly as shown,"* ensuring it focuses on the relevant alphanumeric content.

3.3 System Architecture & Tax Inquiry

The system logic is encapsulated in a Python application (`app.py`).

1. **Input:** The system accepts an image upload or a video stream.
2. **Inference:** The selected pipeline detects the plate, crops it, and passes it to the OCR model.
3. **Normalization:** The raw OCR output is normalized using Regex to remove whitespace and invalid characters.
4. **API Query:** The application performs web scraping to query the Banten Provincial Government's vehicle tax inquiry portal by submitting a POST request with the parsed license plate components (regional prefix, numeric portion, and letter suffix).
5. **Output:** The system displays the retrieved vehicle details in a structured table format. Each field provides specific information about the registered vehicle:

4 Implementation & Results

4.1 Experimental Results

We evaluated two model configurations for license plate detection on Indonesian vehicle images:

Field	Description
NO. POLISI	The official license plate number registered in the database
MEREK	Vehicle manufacturer/brand
TIPE/MODEL	Specific model name of the vehicle
TAHUN / CC / BBM	Manufacturing year, engine displacement (cc), and fuel type
WARNA	Registered vehicle color
TGL. AKHIR PKB	Tax expiry date
JUMLAH	Total tax amount due

Table 1: Output fields from the Banten vehicle tax inquiry system

1. **Pipeline A (CNN-based):** YOLOv8n, a lightweight convolutional neural network optimized for real-time inference.
2. **Pipeline B (Transformer-based):** RT-DETRv2, a Real-Time Detection Transformer that leverages global attention mechanisms for improved spatial understanding.

4.1.1 Quantitative Analysis

We evaluated detection performance using mean Average Precision (mAP) at IoU threshold 0.5, Recall, and inference speed metrics. The results on the validation set are summarized in Table 1.

Model	mAP@0.5	Recall@0.5	FPS	Avg Latency (ms)
Baseline (YOLOv8)	0.759	0.811	**63.2**	**15.8**
Proposed (RT-DETRv2)	**0.899**	**0.971**	25.9	38.6

Table 2: Performance comparison of YOLOv8 vs. RT-DETRv2

RT-DETRv2 achieved a mAP@0.5 of 0.8991, outperforming YOLOv8 by 18.4% and demonstrating a recall of 0.971 compared to YOLOv8's 81.2%. While YOLOv8 maintains a speed advantage at 63.2 FPS compared to RT-DETRv2's 25.9 FPS, the transformer-based model still achieves real-time performance suitable for vehicle tax inquiry systems where detection accuracy is prioritized.

4.2 Training Loss Curves

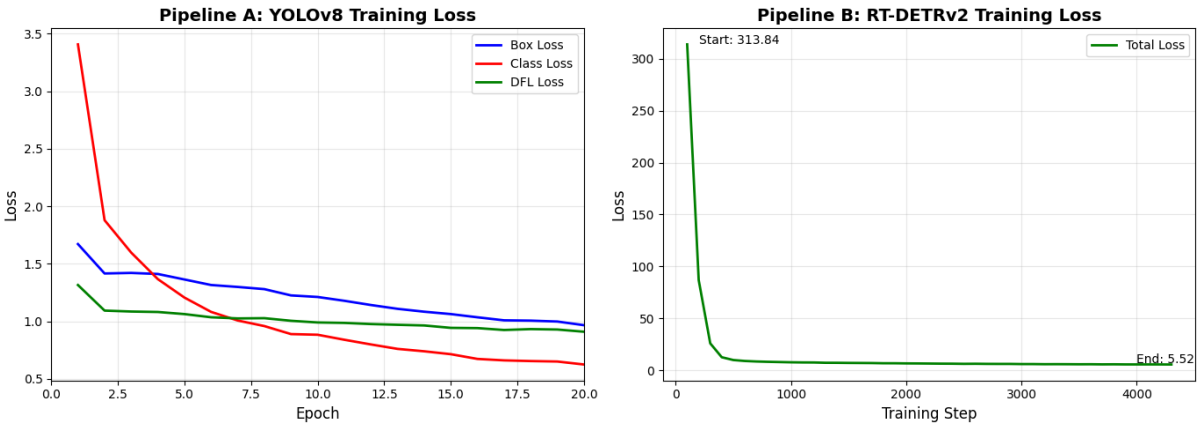

Figure 2: Model Training Graphs

Figure 1 illustrates the training losses of both pipelines. YOLOv8, left, shows consistent losses across all three curves over 20 epochs. The RT-DETRv2 model, right, shows the usual characteristics of a transformer model with a sharp loss reduction from 313.841 down to 7.651 within the first 1,000 training steps, followed by a stable convergence without overfitting down to the final training loss of 5.52. The smooth convergence combined with the strong mAP score of 0.899, indicates that the chosen hyperparameters were appropriate for the use of fine-tuning on the Indonesian License Plate Detection.

5 Discussion & Limitations

5.1 Performance Analysis

The integration of a Vision-Language Model (DeepSeek-OCR) represents a significant trade-off. While it offers superior semantic understanding—capable of correcting common mistakes like confusing '0' for 'D' based on position—it introduces higher latency compared to lightweight CNN-based OCRs. Pipeline A (YOLOv8) typically offers faster initial detection, but Pipeline B (RT-DETRv2) demonstrates higher robustness in dense traffic scenes where plates are partially hidden.

5.2 Challenges and Limitations

- **Computational Cost:** The use of a VLM for OCR requires significant GPU VRAM (approx. 12GB+ for acceptable speeds), making it difficult to deploy on standard edge devices like a Raspberry Pi without heavy quantization.
- **Dependency on External APIs:** The "Tax Inquiry" feature is entirely dependent on the uptime of the Banten Government's server. During testing, intermittent downtime was observed, highlighting the need for a cached local database.
- **Environmental Factors:** Extreme lighting conditions (e.g., direct headlight glare at night or heavy tropical rain) remain a challenge for the optical sensors, degrading the input quality before it reaches the model.

6 Conclusion & Future Work

This project successfully demonstrated a proof-of-concept for a modern, AI-driven vehicle tax compliance system tailored to Indonesian civic infrastructure. By comparing YOLOv8 and RT-DETRv2, we provided empirical evidence on the viability of Transformers in real-time constrained environments, with RT-DETRv2 achieving a 97.1% recall rate, critical for government applications where missed detections lead to lost tax revenue. The integration of DeepSeek-OCR highlights the potential of Generative AI in traditional computer vision tasks, offering high accuracy at the cost of computational resources.

6.1 Benefits to Indonesian Government

The proposed system offers several tangible benefits for Indonesian government agencies:

1. By automating license plate detection at strategic checkpoints (toll gates, parking facilities, government buildings), authorities can identify vehicles with expired PKB (Pajak Kendaraan Bermotor) in real-time, enabling immediate enforcement or notification. With an estimated 39% [13] of vehicles in some regions operating with expired registration, automated detection could significantly increase provincial tax revenue.

2. Scalable National Deployment: While this prototype targets the Banten province database, integration with the national database would enable country wide enforcement, addressing the current challenge of being limited only to the Banten province database.

6.2 Future Work

1. Edge Optimization: Quantizing the DeepSeek-OCR model to 4-bit or using a distilled model to enable deployment on embedded devices like NVIDIA Jetson.
2. Offline Database: Implementing a locally cached tax database with periodic synchronization to remove dependency on external API availability and reduce query latency.
3. Multi-Lane Tracking: Extending the system to track vehicles across multiple lanes using DeepSORT to prevent duplicate queries for the same vehicle.

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A Team Contribution Statement

Member	Contribution
Bryan Anthony (2702283934)	Computer Vision Engineer. Implemented the DeepSeek-OCR fine-tuning pipeline using Unsloth. Trained and validated the YOLOv8 and RT-DETRv2 detection models.
Frederick Ryan Suryadi (2702223033)	Data Engineer & Reporting. Curated and preprocessed the Kaggle dataset. Conducted performance benchmarking (mAP, FPS calculations). Authored the final report, managed GitHub repository structure, and created the system architecture diagrams.
Nico Valerian Marcello (2702250242)	Web App, Reporting & Video Editor. Developed the Streamlit dashboard for the user interface. Handled the backend API integration with Banten tax services and developed the main inference script. Authored the final report. Presentation video editor.

Table 3: Breakdown of Team Responsibilities

B System Snapshots

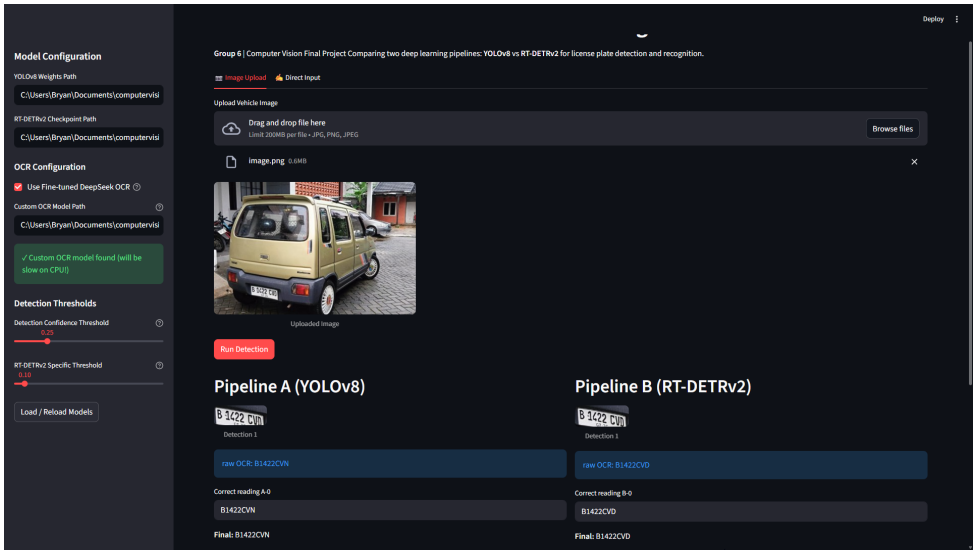

Figure 3: App Interface Showing Detection Results

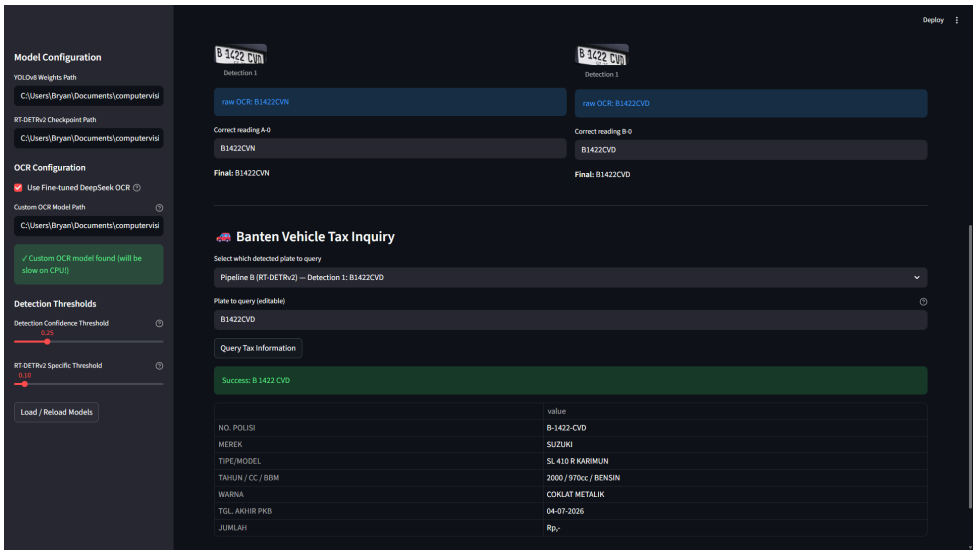

Figure 4: App Interface Showing Tax Query Results From License Plate Detector

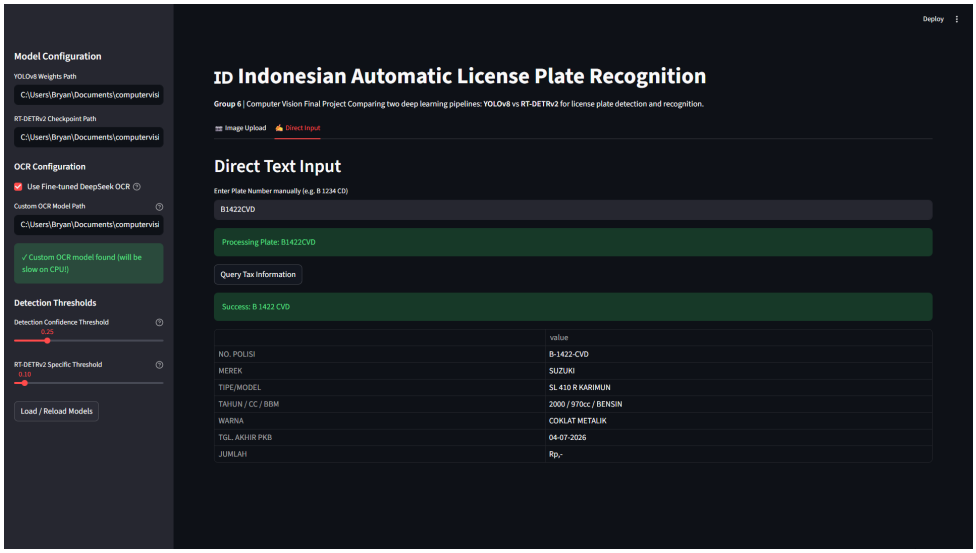

Figure 5: App Interface Showing License Plate Query Results